

# AI2002 – Quiz 3 Solutions

**Included quizzes:** Quiz 3A and Quiz 3B

**Checked and rechecked logic:** the solutions below follow the exact wording and constraints in the uploaded quizzes.

## Quiz 3A Solution

**Scenario:** Four tasks {T1, T2, T3, T4} must be assigned to shifts with domain {Morning (M), Afternoon (A), Night (N)}.

**Constraints:** (T1  $\neq$  T2), (T2 before T3), (T3  $\neq$  T4), (T1 before T4)

**Order:** Morning < Afternoon < Night

### Part 1: Forward Checking

Given **T1 = Afternoon (A)**:

- From **T1  $\neq$  T2**, remove A from T2.
- From **T1 before T4**, T4 must be after A, so T4 = {N}.
- T3 has no direct constraint with T1, so its domain stays unchanged.

| Task | Initial Domain | After T1 = A (Forward Checking) |
|------|----------------|---------------------------------|
| T2   | {M, A, N}      | {M, N}                          |
| T3   | {M, A, N}      | {M, A, N}                       |
| T4   | {M, A, N}      | {N}                             |

### Part 2: MRV & Selection

From Part 1, the domain sizes are:

- T2: size 2  $\rightarrow$  {M, N}
- T3: size 3  $\rightarrow$  {M, A, N}
- T4: size 1  $\rightarrow$  {N}

So MRV selects **T4** because it has the smallest remaining domain.

Assign the earliest possible shift from its remaining domain: **T4 = N**.

Now apply Forward Checking again:

- Constraint **T3  $\neq$  T4** with T4 = N removes N from T3.
- No direct constraint connects T4 to T2 other than what was already handled indirectly through T1, so T2 stays the same.

| Remaining Task | Domain from Part 1 | New Domain (After FC) |
|----------------|--------------------|-----------------------|
| T2             | {M, N}             | {M, N}                |

|    |           |        |
|----|-----------|--------|
| T3 | {M, A, N} | {M, A} |
|----|-----------|--------|

**Does this branch result in a failure? No**

Reason: no domain becomes empty after assigning  $T4 = N$ .

### Part 3: Constraint Inconsistency

**Given:** Tasks {X, Y, Z}, Domains {Morning, Night}, Constraints (X before Y), (Y before Z), (X after Z)

*With only two ordered values, Morning < Night.*

**1. If X = Morning, what are the domains of Y and Z using Forward Checking?**

- From **X before Y** and X = Morning, Y must be after Morning, so **Y = {Night}**.
- From **X after Z** and X = Morning, Z must be before Morning. But there is no value before Morning in {Morning, Night}, so **Z = {}**.

**D(Y) = {Night}    D(Z) = {}**

**2. Apply Arc Consistency specifically to the edge X after Z. Does it eliminate any values from X or Z?**

**Result:** Yes. The constraint X after Z means Z must be earlier than X. Since the only ordered pair available is (Z = Morning, X = Night), Arc Consistency removes **Morning** from X and removes **Night** from Z. So AC prunes the domains to **X = {Night}** and **Z = {Morning}**.

**3. Is this problem solvable? Briefly explain why based on your domain results.**

**Answer:** No, it is not solvable. After AC, we get X = Night and Z = Morning. Then the constraint X before Y forces Y to be after Night, which is impossible in the domain {Morning, Night}. Also Y before Z would require Y to be before Morning, which is also impossible. Therefore the constraint set is inconsistent.

## Quiz 3B Solution

**Scenario:** Four milestones {M1, M2, M3, M4} must be assigned to phases with domain {Phase 1 (P1), Phase 2 (P2), Phase 3 (P3)}.

**Constraints:** (M1 before M2), (M2  $\neq$  M3), (M3 before M4), (M1  $\neq$  M3)

*Order:  $P1 < P2 < P3$*

### Part 1: Forward Checking

Given **M1 = P2**:

- From **M1 before M2**, M2 must be after P2, so  $M2 = \{P3\}$ .
- From **M1  $\neq$  M3**, remove P2 from M3, so  $M3 = \{P1, P3\}$ .
- M4 has no direct constraint with M1, so it stays unchanged.

| Milestone | Initial Domain | After M1 = P2 (Forward Checking) |
|-----------|----------------|----------------------------------|
| M2        | {P1, P2, P3}   | {P3}                             |
| M3        | {P1, P2, P3}   | {P1, P3}                         |
| M4        | {P1, P2, P3}   | {P1, P2, P3}                     |

### Part 2: MRV & Selection

From Part 1, the domain sizes are:

- M2: size 1  $\rightarrow$  {P3}
- M3: size 2  $\rightarrow$  {P1, P3}
- M4: size 3  $\rightarrow$  {P1, P2, P3}

So MRV selects **M2**.

Assign the earliest possible phase from its remaining domain: **M2 = P3**.

Apply Forward Checking again:

- From **M2  $\neq$  M3**, remove P3 from M3, so M3 becomes **{P1}**.
- M4 is not directly constrained by M2, so it remains **{P1, P2, P3}**.

| Remaining Milestone | Domain from Part 1 | New Domain (After FC) |
|---------------------|--------------------|-----------------------|
| M3                  | {P1, P3}           | {P1}                  |
| M4                  | {P1, P2, P3}       | {P1, P2, P3}          |

**Does this branch result in a failure? No**

Reason: no domain becomes empty after assigning  $M2 = P3$ .

### Part 3: Forward Checking vs. Arc Consistency

**Given:** Variables {X, Y, Z}, Domains {P1, P2, P3}, Constraints (X before Y), (Y before Z)

**1. If we assign  $X = P1$ , what are the domains of  $Y$  and  $Z$  using Forward Checking?**

- From  **$X$  before  $Y$**  and  $X = P1$ ,  $Y$  must be after  $P1$ , so  $Y = \{P2, P3\}$ .
- $Z$  has no direct constraint with  $X$ , so it remains  $\{P1, P2, P3\}$ .

$D(Y) = \{P2, P3\}$      $D(Z) = \{P1, P2, P3\}$

**2. Now apply Arc Consistency to the constraint ( $Y$  before  $Z$ ). Does it remove any additional value from the domain of  $Y$ ?**

**Result:** Yes. Under the constraint  $Y$  before  $Z$ , each value of  $Y$  must have some later supporting value in  $Z$ . Value  **$P3$**  in  $Y$  has no supporting value in  $Z$  because nothing comes after  $P3$ . Therefore AC removes  $P3$  from  $Y$ , leaving  $Y = \{P2\}$ .

**3. Which method gives a smaller domain for  $Y$  after propagation?**

**Answer:** Arc Consistency gives a smaller domain. Forward Checking leaves  $Y = \{P2, P3\}$ , while Arc Consistency further prunes it to  $Y = \{P2\}$ .

**Final verification summary**

| Quiz      | Key checked point                                        | Verified conclusion |
|-----------|----------------------------------------------------------|---------------------|
| 3A Part 1 | $T1 = A$ affects $T2$ and $T4$ only                      | Correct             |
| 3A Part 2 | MRV selects $T4 = N$                                     | Correct             |
| 3A Part 3 | $X = \text{Morning}$ makes $Z$ empty under $X$ after $Z$ | Correct             |
| 3B Part 1 | $M1 = P2$ gives $M2 = \{P3\}$ , $M3 = \{P1, P3\}$        | Correct             |
| 3B Part 2 | MRV selects $M2 = P3$ , then $M3 = \{P1\}$               | Correct             |
| 3B Part 3 | AC removes $P3$ from $Y$                                 | Correct             |